

L03 – Balanced (Single-phase Equivalent) Power Flow

Foundations Lecture Series

Course Instructor

Microgrid 3-phase Security AI Project

Foundations Lecture 3

Keywords: positive sequence, single-line diagram, three-phase symmetry, when balance holds

Lecture roadmap

- ① What “balanced” actually means
- ② The single-phase equivalent: how we collapse three phases into one
- ③ Positive-sequence networks
- ④ When the balanced assumption is appropriate
- ⑤ When it lies: preview of L04

Prerequisite

L01 – you should be comfortable with Y_{bus} and the PF equations. L02 helpful but not required.

Definition of balanced operation

TODO: *Three-phase voltages and currents equal in magnitude, 120° apart, sinusoidal. Source is symmetric, loads are symmetric, line parameters are symmetric. Visual: phasor diagram of a balanced set.*

Symmetry assumptions in three places

TODO: (i) Source symmetry, (ii) load symmetry, (iii) line geometry symmetry (or transposition). All three usually hold in transmission, often fail in distribution. Bridge: under all three, three coupled equations collapse to one.

Collapsing three phases into one

TODO: Show that under balance, $V_b = a^2 V_a$ and $V_c = a V_a$ with $a = e^{j2\pi/3}$. Therefore we only need to solve for phase a – the other phases follow by rotation. This is the “single-line diagram” you usually see.

Single-line diagrams in practice

TODO: *Visual example: a small system shown both as a true three-phase circuit and as its single-line equivalent. Note that one-line diagrams in textbooks and pandapower's `runpp` both implicitly use this collapse.*

Positive sequence in 30 seconds

TODO: *Forward-reference to L04's symmetrical components: positive, negative, zero. Under balance, only positive sequence is non-zero. Negative and zero sequence appear only when balance breaks.*

The math is L01's math

TODO: *Show that the PF equations from L01 are the positive-sequence PF equations. `pp.runpp` solves them. Nothing new computationally – only the interpretation changes.*

When the assumption is fine

TODO: *Transmission grids: three-phase loads, transposed lines, large industrial loads. The assumption introduces small ($< 1\%$) errors. List several practical examples where balanced PF is the right tool.*

Quick-and-dirty hosting analyses

TODO: *For distribution-level studies where the question is “rough voltage profile, rough loading”, balanced PF is sometimes acceptable as a first pass. Note caveats: never claim phase-specific results from a balanced model.*

Where the balanced assumption breaks

TODO: *Single-phase loads, single-phase rooftop PV, EV charging, asymmetric line geometry, fault analysis. Show a small example where balanced PF reports “all good” but the actual three-phase analysis (L04) shows phase A undervoltage.*

Diagnostic: when do I need L04?

TODO: *A short checklist for students: (1) Are loads single-phase? (2) Is PV single-phase? (3) Are line parameters asymmetric? (4) Is the network LV? If any yes, you need unbalanced PF.*

Homework

TODO: Take the L01 three-bus system, intentionally make one of the loads single-phase, and explain why `runpp` silently lies. Run `runpp_3ph` on the same system and report the per-phase voltage spread.

Exit checklist

TODO: *Items: (1) I can state the three symmetry assumptions, (2) I can draw a one-line from a three-phase, (3) I can decide whether to use `runpp` or `runpp_3ph`, (4) I know the connection to symmetrical components is coming in L04.*

TODO: *Grainger & Stevenson chapter on per-phase analysis, IEC/IEEE definitions of balanced operation, pandapower “AC Power Flow” page.*